5.2.3 Transformations

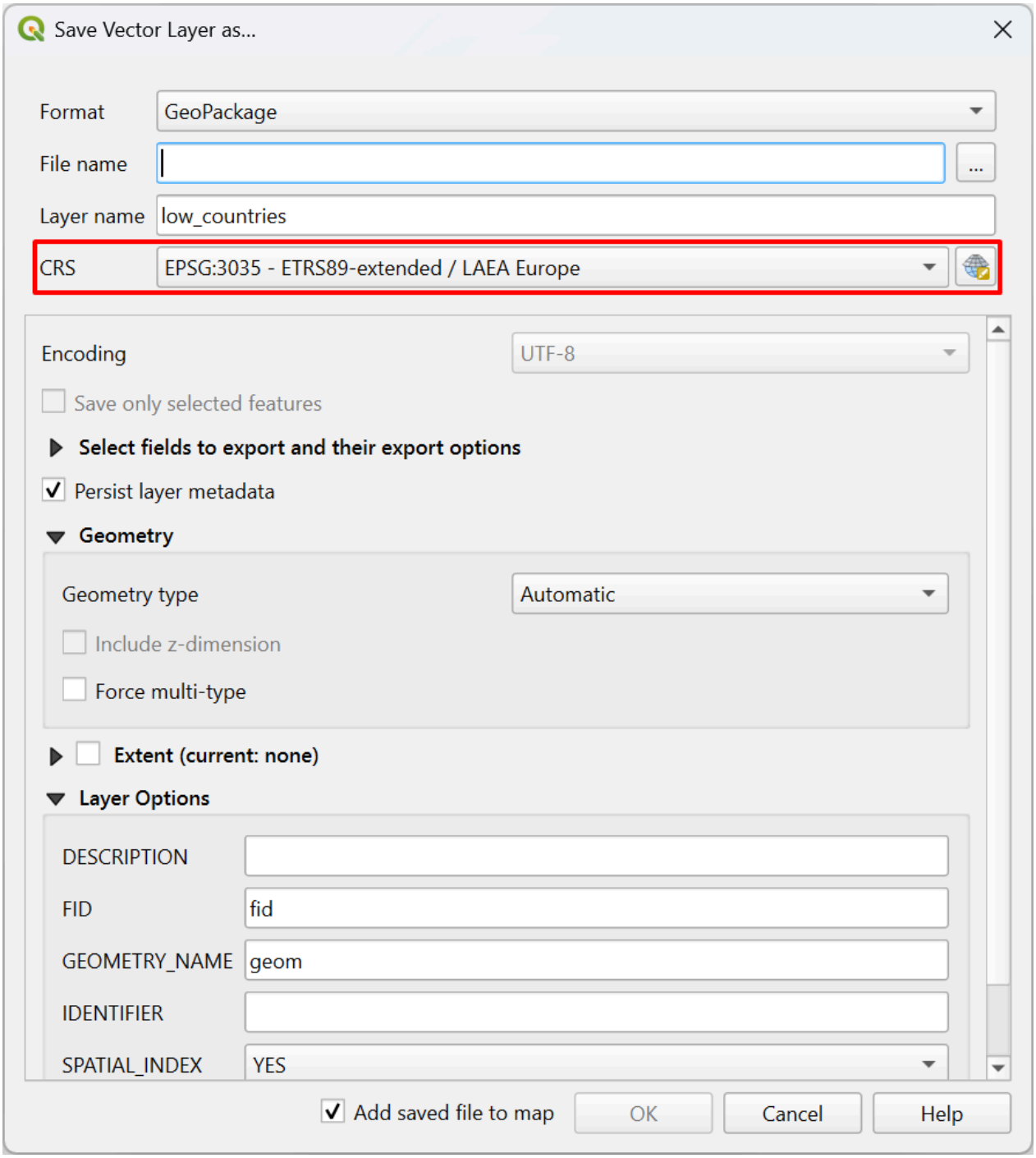
Transformations

Coordinates in one coordinate reference system can be converted with a mathematical formula into coordinates in another CRS. When this involves changing the map projection, the process is commonly referred to as *reprojection* in QGIS. More generally, it is called a **coordinate transformation**.

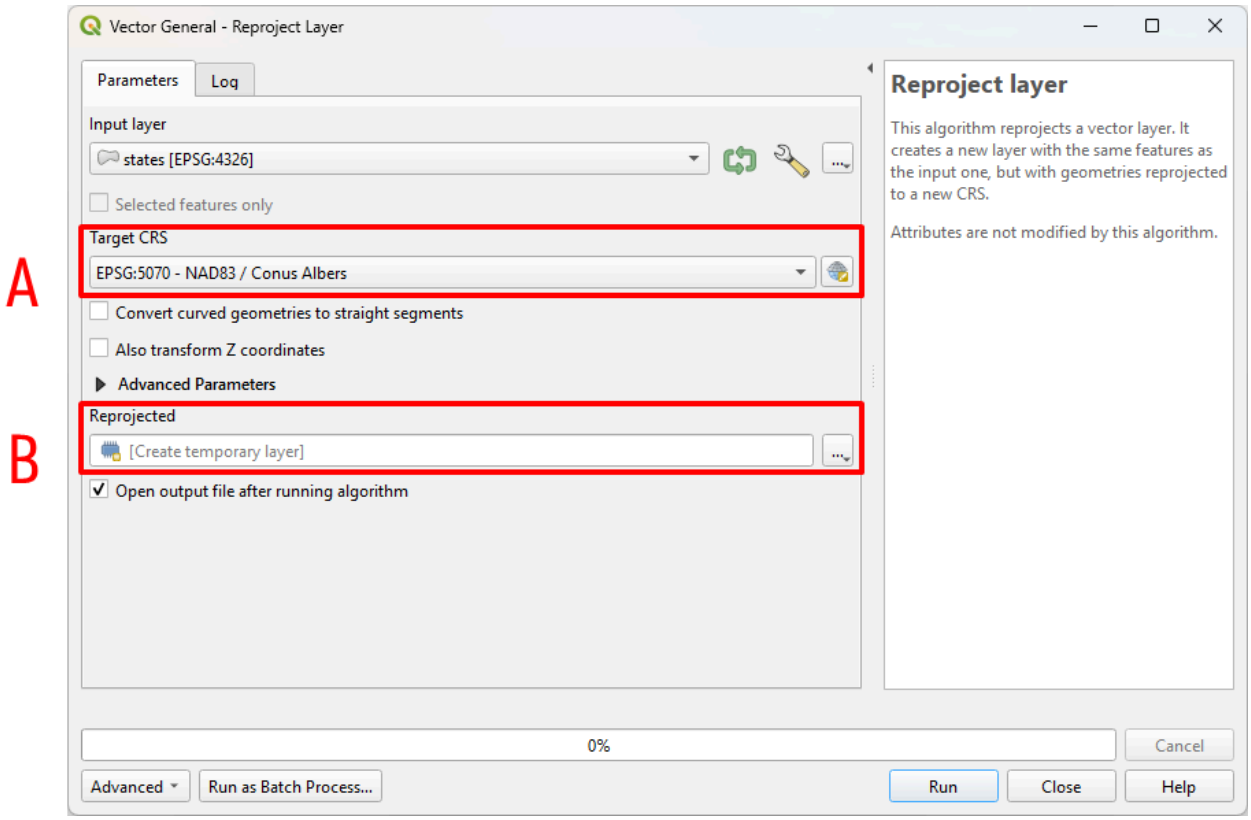
Explicit transformations

When you explicitly reproject a layer using the methods described below, QGIS creates a new layer with the transformed coordinates. The original layer remains unchanged.

A first common way is to **export the layer to a new layer** with the new CRS. Right-click the layer in the *Layers* panel and choose *Export* ▸ *Save Features As…*. In the dialog, keep *GeoPackage* as *Format* (the default), enter a file name and choose the new CRS under *CRS*.

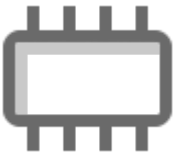


Alternatively, you can use the reprojection tools in the *Processing Toolbox* (*Processing* ▸ *Toolbox*). For vector layers, use *Vector general* ▸ *Reproject layer*. For raster layers, use *GDAL* ▸ *Raster projections* ▸ *Warp (reproject)*. You can also type the tool name in the search box at the top of the toolbox. In the tool, you choose the new CRS under *Target CRS* (A) and, under *Reprojected* (B), whether the result is stored as a temporary layer or as a file on your computer.



Warning

By default, a processing tool such as *Reproject layer* creates a **temporary layer**: QGIS does not write a file, but keeps the layer in the memory of your work session. If you close your project and open it again, the content of temporary layers is no longer available.

In the *Layers* panel, a temporary layer is marked with this icon:  . To save it as a file after all, right-click the layer and choose *Make Permanent...*.

Implicit transformations

Besides these explicit transformations, QGIS also uses **transformations implicitly**. When you load data whose coordinate system differs from the project CRS, QGIS transforms the coordinates on the fly so that the data can be positioned and displayed correctly in the map canvas. This process runs in the background: the original data are not changed, but the coordinates are recalculated for display. Thanks to this, you can load data sources into QGIS and display them together, whatever their CRS.

Transformations can lead to deviations

Coordinate transformations are **not always exact**. Depending on the transformation used and the accuracy of its parameters, they can introduce small positional errors, sometimes amounting to several metres or more.

Several mathematical transformations may be available for converting coordinates between two CRSs, each with its own accuracy. In that case, QGIS asks you which one to use and shows the accuracy of each. Choose the most accurate transformation that is suitable for your area.

Within one project, always use the same transformation for the same pair of CRSs. Otherwise, identical features may no longer overlap exactly. Also avoid unnecessary transformations: every transformation can introduce small inaccuracies.

Coordinate systems are not simply interchangeable!

Although you do not need to transform geodata explicitly to visualise them in the map canvas, it is in certain cases **advisable to do so**.

CRS depending on the analysis

Spatial analysis tools often **use the original data (and therefore the original coordinates)**, so the CRS of each layer affects the result:

- Geographic coordinate systems are generally unsuitable for Cartesian calculations. Some QGIS tools can produce misleading results when used with geographic coordinates: a buffer distance, for example, is then expressed in degrees rather than metres (see Vector processing tools).
- Not all coordinate systems are suitable for all regions. A CRS that gives reliable results for one region can show large *distortions* for other regions. These distortions carry over into the distances, areas and shapes the software calculates for this layer, and so lead to wrong results.
- The suitable CRS also depends on the type of analysis. If you want to calculate areas for the features in a layer, a layer that uses an equal-area CRS will give reliable results; other CRSs can distort areas, especially over large regions.
- If you want to run a spatial analysis on several layers (for example, calculating the overlap between two layers), using **the same suitable CRS** for all layers can help avoid inconsistencies and improve processing efficiency.

Why transform data explicitly?

You learned that if a layer does not use the same CRS as the project, QGIS transforms it on the fly to display it in the map canvas. This allows layers with different CRSs to be displayed together, but coordinate transformations can introduce small positional inaccuracies. As a result, the displayed position of a feature may differ slightly from its position in the original data. In addition, many analysis tools work with the coordinates stored in the layer, not with the coordinates as they appear in the map canvas. Transforming your data explicitly to a suitable common CRS can help ensure consistency, particularly when performing spatial analyses.